Preceding material.

$2^\circ)$ $Z_{\rho,\sigma},\ \gamma_{\rho,\sigma}$.

Let

$$
\begin{array}{l}
Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu \\
\qquad \| \text{ dfn} \\
\left\{ (u, \mu, \varepsilon_2, \varepsilon_3) \;\middle|\; \begin{array}{l} u \text{ normalizes } \mathcal{G}^\natural \text{ and induces the identity in } \mathcal{G}^+/\mathcal{G}^\natural \\ a)\ \ \exists \alpha \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\alpha) = \varepsilon_3, \quad u(\rho) = \alpha(\rho) \\ b)\ \ \exists \beta \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\beta) = \varepsilon_2, \quad u(\sigma) = \beta(\sigma) \\ c)\ \ u(\varepsilon_0) = \varepsilon_0^\mu \\ d)\ \ u \text{ normalizes } \mathcal{G}^{\natural|\sharp} \text{ and induces the identity on } (\mathcal{G}^+/\mathcal{G}^{\natural|\sharp})/\operatorname{Im}\Pi \end{array} \right\}.
\end{array} \tag{4}
$$

Preceding material.

$2^\circ)$ $Z_{\rho,\sigma},\ \gamma_{\rho,\sigma}$.

Let

$$
\begin{array}{l}
Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu \\
\qquad \| \, \mathrm{dfn} \\
\left\{ (u, \mu, \varepsilon_2, \varepsilon_3) \;\middle|\; \begin{array}{ll}
a) & \exists \alpha \in \mathcal{G}_\tau^{\natural|\sharp}, \text{ with } \varepsilon_{\mathcal{G}}(\alpha) = \varepsilon_3, \quad u(\rho) = \alpha(\rho) \\
b) & \exists \beta \in \mathcal{G}_\tau^{\natural|\sharp}, \text{ with } \varepsilon_{\mathcal{G}}(\beta) = \varepsilon_2, \quad u(\sigma) = \beta(\sigma) \\
c) & u(\varepsilon_0) = \varepsilon_0^{\mu} \\
d) & u \text{ normalizes } \mathcal{G}^{\natural|\sharp} \text{ and induces the identity on } (\mathcal{G}^+/\mathcal{G}^{\natural|\sharp})/\operatorname{Im}\Pi
\end{array} \right\}.
\end{array} \tag{4}
$$

Preceding material.

$2^\circ)$ $Z_{\rho,\sigma},\ \gamma_{\rho,\sigma}$.

Let

$$
\begin{array}{l}
Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu \\
\qquad \| \, \mathrm{dfn} \\
\left\{ (u, \mu, \varepsilon_2, \varepsilon_3) \;\middle|\; \begin{array}{l} \dagger \\ a)\ \ \exists \alpha \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\alpha) = \varepsilon_3, \quad u(\rho) = \alpha(\rho) \\ b)\ \ \exists \beta \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\beta) = \varepsilon_2, \quad u(\sigma) = \beta(\sigma) \\ c)\ \ u(\varepsilon_0) = \varepsilon_0^{\mu} \\ d)\ \ u \text{ normalizes } \mathcal{G}^{\natural|\sharp} \text{ and induces the identity on } (\mathcal{G}^+/\mathcal{G}^{\natural|\sharp})/\operatorname{Im}\Pi \end{array} \right\}.
\end{array} \tag{4}
$$

Preceding material.

2°) $Z_{\rho,\sigma},\ \gamma_{\rho,\sigma}$.

Let

$$Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu$$
$$\| \text{ dfn}$$
$$(4) \qquad \left\{ (u,\mu,\varepsilon_2,\varepsilon_3) \;\middle|\; \begin{array}{l} u \text{ normalizes } \mathcal{G}^\natural \text{ and induces the identity in } \mathcal{G}^+/\mathcal{G}^\natural \\ a)\ \ \exists \alpha \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\alpha) = \varepsilon_3, \quad u(\rho) = \alpha(\rho) \\ b)\ \ \exists \beta \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\beta) = \varepsilon_2, \quad u(\sigma) = \beta(\sigma) \\ c)\ \ u(\varepsilon_0) = \varepsilon_0^\mu \\ d)\ \ u \text{ normalizes } \mathcal{G}^{\natural|\sharp} \text{ and induces the identity on } (\mathcal{G}^+/\mathcal{G}^{\natural|\sharp})/\operatorname{Im}\Pi \end{array} \right\}.$$

Preceding material.

2°) $Z_{\rho,\sigma},\ \gamma_{\rho,\sigma}$.

Let

$$
\begin{array}{l}
Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu \\
\qquad \| \,\mathrm{dfn} \\
\left\{ (u, \mu, \varepsilon_2, \varepsilon_3) \;\middle|\; \begin{array}{ll}
a) & \exists \alpha \in \mathcal{G}_\tau^{\natural\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\alpha) = \varepsilon_3, \quad u(\rho) = \alpha(\rho) \\
b) & \exists \beta \in \mathcal{G}_\tau^{\natural\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\beta) = \varepsilon_2, \quad u(\sigma) = \beta(\sigma) \\
c) & u(\varepsilon_0) = \varepsilon_0^{\mu} \\
d) & u \text{ normalizes } \mathcal{G}^{\natural\sharp} \text{ and induces the identity on } (\mathcal{G}^+/\mathcal{G}^{\natural\sharp})/\operatorname{Im}\Pi
\end{array} \right\}.
\end{array} \tag{4}
$$

Preceding material.

2°) $Z_{\rho,\sigma},\ \gamma_{\rho,\sigma}$.

Let

$$Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu$$
$$\| \text{ dfn}$$
$$(4) \qquad \left\{ (u, \mu, \varepsilon_2, \varepsilon_3) \;\middle|\; \begin{array}{l} \dagger \\ a)\ \ \exists \alpha \in \mathcal{G}_\tau^{\natural\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\alpha) = \varepsilon_3, \quad u(\rho) = \alpha(\rho) \\ b)\ \ \exists \beta \in \mathcal{G}_\tau^{\natural\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\beta) = \varepsilon_2, \quad u(\sigma) = \beta(\sigma) \\ c)\ \ u(\varepsilon_0) = \varepsilon_0^\mu \\ d)\ \ u \text{ normalizes } \mathcal{G}^{\natural\sharp} \text{ and induces the identity on } (\mathcal{G}^+/\mathcal{G}^{\natural\sharp})/\operatorname{Im}\Pi \end{array} \right\}.$$

Preceding material.

2°) $Z_{\rho,\sigma},\ \gamma_{\rho,\sigma}$.

Let

$$Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu$$
$$\| \text{ dfn}$$
$$(4) \qquad \left\{ (u,\mu,\varepsilon_2,\varepsilon_3) \;\middle|\; \begin{array}{l} u \text{ normalizes } \mathcal{G}^\natural \text{ and induces the identity in } \mathcal{G}^+/\mathcal{G}^\natural \\ a)\ \ \exists \alpha \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\alpha) = \varepsilon_3, \quad u(\rho) = \alpha(\rho) \\ b)\ \ \exists \beta \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\beta) = \varepsilon_2, \quad u(\sigma) = \beta(\sigma) \\ c)\ \ u(\varepsilon_0) = \varepsilon_0^\mu \\ d)\ \ u \text{ normalizes } \mathcal{G}^{\natural|\sharp} \text{ and induces the identity on } (\mathcal{G}^+/\mathcal{G}^{\natural|\sharp})/\operatorname{Im}\Pi \end{array} \right\}.$$

Preceding material.

2°) $Z_{\rho,\sigma},\ \gamma_{\rho,\sigma}$.

Let

$$Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu$$
$$\| \text{ dfn}$$

$$(4) \qquad \left\{ (u,\mu,\varepsilon_2,\varepsilon_3) \;\middle|\; \begin{array}{ll} a) & \exists \alpha \in \mathcal{G}_\tau^{\natural\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\alpha) = \varepsilon_3, \quad u(\rho) = \alpha(\rho) \\ b) & \exists \beta \in \mathcal{G}_\tau^{\natural\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\beta) = \varepsilon_2, \quad u(\sigma) = \beta(\sigma) \\ c) & u(\varepsilon_0) = \varepsilon_0^\mu \\ d) & u \text{ normalizes } \mathcal{G}^{\natural\sharp} \text{ and induces the identity on } (\mathcal{G}^+/\mathcal{G}^{\natural\sharp})/\operatorname{Im}\Pi \end{array} \right\}.$$

Preceding material.

2°) $Z_{\rho,\sigma},\ \gamma_{\rho,\sigma}$.

Let

$$Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu$$
$$\| \text{ dfn}$$

$$(4) \qquad \left\{ (u,\mu,\varepsilon_2,\varepsilon_3) \;\middle|\; \begin{array}{l} \dagger \\ a)\ \ \exists \alpha \in \mathcal{G}_\tau^{\natural\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\alpha) = \varepsilon_3, \quad u(\rho) = \alpha(\rho) \\ b)\ \ \exists \beta \in \mathcal{G}_\tau^{\natural\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\beta) = \varepsilon_2, \quad u(\sigma) = \beta(\sigma) \\ c)\ \ u(\varepsilon_0) = \varepsilon_0^\mu \\ d)\ \ u \text{ normalizes } \mathcal{G}^{\natural\sharp} \text{ and induces the identity on } (\mathcal{G}^+/\mathcal{G}^{\natural\sharp})/\operatorname{Im}\Pi \end{array} \right\}.$$